REVENUE, PROFIT & COMMISSION TRACKER

Full Project Documentation

Data Analysis Project · Microsoft Excel Dashboard · Pool Supply Sales

171 **Transactions**	**$17,110.60** **Total Sales**	**$6,356.70** **Total Profit**	**37.15%** **Profit Margin**

10 Products · 6 States · 4 Sales Reps · 12 Months

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1. Project Overview

This project is a fully interactive **Microsoft Excel dashboard** built to track and analyse revenue, profit, and sales commissions for a pool supply business. It consolidates **171 sales transactions** spanning a full calendar year across **6 US states**, **10 products**, and **4 sales representatives** into a single visual analytics interface driven by PivotTables, PivotCharts, and Slicers.

1.1 Objectives

- Track total revenue, cost, and profit on a monthly and annual basis
- Measure sales representative performance via commission tracking
- Identify top-performing products by revenue and profit contribution
- Analyse geographic sales distribution across six US states
- Provide an interactive, filter-driven dashboard for business decision-making

1.2 Key Performance Indicators

KPI	Value	Formula Used
Total Sales Revenue	$17,110.60	SUM(Sale Price column)
Total Cost Price	$10,753.90	SUM(Store Cost column)
Total Profit	$6,356.70	SUM(Profit column)
Profit Margin	37.15%	Total Profit / Total Sales × 100
Average Transaction	$100.06	AVERAGE(Sale Price column)
Total Transactions	171	COUNT(Transaction Number)

2. Repository & File Structure

File	Type	Role	Size (approx.)
Data_Set.xlsx	Excel (.xlsx)	Raw transactional dataset — source of truth	~50 KB
Profit_Tracker.xlsx	Excel (.xlsx)	Interactive dashboard with Pivot Charts & Slicers	~120 KB
Dashboard_Screenshot.png	Image (.png)	Visual preview for GitHub README	~200 KB
README.docx	Word (.docx)	GitHub README with project summary and usage guide	~80 KB
Documentation.pdf	PDF (.pdf)	This full documentation file	~300 KB

2.1 Recommended GitHub Repository Structure

```
revenue-profit-commission-tracker/ ■■■ data/ ■ ■■■ Data_Set.xlsx ← Raw dataset (source
of truth) ■■■ dashboard/ ■ ■■■ Profit_Tracker.xlsx ← Interactive Excel dashboard ■■■
```

```
docs/ ■ ■■■ Documentation.pdf ← Full project documentation ■ ■■■
Dashboard_Screenshot.png ■■■ README.docx ← GitHub README ■■■ LICENSE ← MIT License
```

3. Dataset Description & Schema

The raw dataset (**Data_Set.xlsx**) contains one row per sales transaction. There are **171 rows** of data plus one header row, giving 172 total rows. The dataset has **11 columns** covering transaction metadata, pricing, profit, commission, sales rep identity, and geographic location.

3.1 Column Schema

#	Column Name	Data Type	Example	Description
1	Month	Text	Jan	Abbreviated month name (Jan–Dec)
2	Transaction Number	Integer	1001	Unique transaction ID starting at 1001, increments by 1
3	Product Code	Integer	8722	Numeric SKU identifying the product
4	Product Description	Text	Water Pump	Human-readable product name
5	Store Cost	Currency	$344.00	Wholesale cost paid by the store per unit
6	Sale Price	Currency	$502.00	Retail price charged to the customer
7	Profit	Currency	$158.00	Gross profit per transaction (Sale Price – Store Cost)
8	Commission 10%	Currency	$50.20	10% of Sale Price paid to the sales representative
9	First Name	Text	Doug	Sales representative first name
10	Last Name	Text	Smith	Sales representative last name
11	Sale Location	Text	AZ	US state abbreviation where the sale occurred

3.2 Products Catalog

Product Description	Code	Store Cost	Sale Price	Unit Profit	Unit Margin
1 Gal Muratic Acid	9212	$4.00	$7.00	$3.00	42.9%
5 Gal Chlorine	6622	$42.00	$77.00	$35.00	45.5%
8 ft Hose	2499	$6.20	$9.20	$3.00	32.6%
Algea Killer 8 oz	6119	$9.00	$14.00	$5.00	35.7%
AutoVac	2242	$60.00	$124.00	$64.00	51.6%
Chlorine Test Kit	1109	$3.00	$8.00	$5.00	62.5%
Net	2877	$11.40	$16.30	$4.90	30.1%

Product Description	Code	Store Cost	Sale Price	Unit Profit	Unit Margin
Pool Cover	9822	$58.30	$98.40	$40.10	40.8%
Skimmer	4421	$45.00	$87.00	$42.00	48.3%
Water Pump	8722	$344.00	$502.00	$158.00	31.5%

3.3 Sales Representatives & Territories

Sales Rep	States Covered	Commission Rate
Charlie Barns	AZ, CA, NM, NV	10% of Sale Price
Juan Hernandez	AZ, CA, CO, NM, NV, UT	10% of Sale Price
Doug Smith	AZ, CA, CO, NM, NV, UT	10% of Sale Price
Hellen Johnson	AZ, CA, NM, NV, UT	10% of Sale Price

4. Data Cleaning Process

Before building the dashboard, the raw dataset was inspected and cleaned to ensure data quality and analytical accuracy. The following steps were performed in sequence inside Excel:

Step 1 — Initial Data Inspection

- Opened Data_Set.xlsx and reviewed all 171 rows for visual anomalies
- Confirmed all 11 column headers were present and correctly named
- Checked the data types for each column using the Format Cells dialog
- Verified the Transaction Number column started at 1001 and was sequential with no gaps
- Confirmed the Month column covered all 12 months (Jan–Dec) with no missing months

Step 2 — Checking for Duplicate Records

Duplicate transactions were checked using Excel's built-in Remove Duplicates feature and a COUNTIF formula:

```
=COUNTIF($B$2:$B$172, B2)
```

Any Transaction Number with a COUNTIF result greater than 1 would indicate a duplicate. **Result: No duplicates found.** All 171 transaction numbers were unique.

Step 3 — Handling Missing Values

Each column was scanned for blank or null values using COUNTBLANK:

```
=COUNTBLANK(A2:A172) ← applied to each column in turn
```

Result: Zero missing values found across all 11 columns. The dataset was complete with no empty cells.

Step 4 — Data Type Validation

Column	Expected Type	Validation Method	Result
Month	Text	Filter dropdown — confirmed only Jan–Dec values	PASS
Transaction Number	Integer	MIN/MAX check: MIN=1001, MAX=1171	PASS
Product Code	Integer	ISNUMBER() check on all cells	PASS
Store Cost	Number	FORMAT check — currency, 2 decimal places	PASS
Sale Price	Number	FORMAT check — currency, 2 decimal places	PASS
Profit	Number	Verified = Sale Price - Store Cost for each row	PASS
Commission 10%	Number	Verified = Sale Price × 10% for each row	PASS
Sale Location	Text	Filter: only AZ, CA, CO, NM, NV, UT present	PASS

Step 5 — Profit Formula Verification

To validate that the Profit column values were correctly calculated, a helper verification column was added temporarily:

```
=F2-E2 ← Sale Price (col F) minus Store Cost (col E)
```

All 171 rows matched. The helper column was then deleted after verification.

Step 6 — Commission Formula Verification

```
=F2*0.10 ← 10% of Sale Price (col F)
```

All commission values matched the 10% rule. **Note:** Minor floating-point imprecision (e.g. $4.9000000000000004 instead of $4.90) was observed due to Excel's binary floating-point representation. This was resolved by applying the ROUND formula to the commission column:

```
=ROUND(F2*0.10, 2)
```

Step 7 — Standardising Text Columns

- Month column: confirmed consistent abbreviated format (Jan, Feb, … Dec) — no variations like 'January' or 'jan'
- Sale Location: confirmed only 2-letter uppercase state codes — no full state names or lowercase entries
- Product Description: confirmed consistent casing and spelling across all 171 rows
- Sales rep names: confirmed consistent capitalisation (First Last format)

Step 8 — Sorting & Structuring for Pivot Tables

- Data was sorted by Month (chronological order: Jan–Dec) then by Transaction Number
- The data range was converted to a formal Excel Table (Insert → Table) to enable dynamic range expansion
- The table was named 'SalesData' for clean formula references
- Header row was locked (freeze top row: View → Freeze Panes → Freeze Top Row)

Note: *The dataset required minimal cleaning — it was already well-structured. The most significant correction was applying ROUND() to address floating-point precision in the Commission column.*

5. Formulas & Calculations

All core metrics in this project are calculated using standard Excel formulas. Below is a complete reference of every formula used, with column references matching the Data_Set.xlsx layout (columns A–K, data rows 2–172).

5.1 Core Transaction Formulas (Data_Set.xlsx — Column G & H)

Column	Formula	Description
G (Profit)	=F2-E2	Gross profit per transaction = Sale Price minus Store Cost
H (Commission 10%)	=ROUND(F2*0.1,2)	Commission earned = 10% of Sale Price, rounded to 2 dp

5.2 KPI Summary Formulas (Profit_Tracker.xlsx — Summary Sheet)

KPI	Excel Formula	Result
Total Sales Revenue	=SUM(SalesData[Sale Price])	$17,110.60
Total Cost Price	=SUM(SalesData[Store Cost])	$10,753.90
Total Profit	=SUM(SalesData[Profit])	$6,356.70
Profit Margin %	=SUM(SalesData[Profit])/SUM(SalesData[Sale Price])	37.15%
Average Sale	=AVERAGE(SalesData[Sale Price])	$100.06
Transaction Count	=COUNTA(SalesData[Transaction Number])	171
Total Commission Paid	=SUM(SalesData[Commision 10%])	$1,711.06

5.3 Profit Margin Formula — Explained

```
Profit Margin = (Total Profit / Total Revenue) × 100 In Excel: =SUM(SalesData[Profit]) / SUM(SalesData[Sale Price]) = 6356.70 / 17110.60 = 0.3715... → Formatted as Percentage: 37.15%
```

5.4 SUMIF Formulas — Product-Level Analysis

To calculate profit and revenue per product outside of PivotTables, SUMIF was used:

```
=SUMIF(SalesData[Product Description], "Water Pump", SalesData[Profit]) → Returns: $3,160.00 =SUMIF(SalesData[Product Description], "Water Pump", SalesData[Sale Price]) → Returns: $10,040.00 General pattern: =SUMIF(SalesData[Product Description], [Product Name], SalesData[Profit])
```

5.5 SUMIF Formulas — Geographic Analysis

```
=SUMIF(SalesData[Sale Location], "AZ", SalesData[Profit]) → Returns: $1,979.10 (Arizona profit) =SUMIF(SalesData[Sale Location], "CA", SalesData[Sale Price]) → Returns: Total California revenue General pattern: =SUMIF(SalesData[Sale Location], [State Code], SalesData[Profit])
```

5.6 SUMIF Formulas — Monthly Analysis

```
=SUMIF(SalesData[Month], "Jan", SalesData[Sale Price]) → Returns: $1,409.50 (January total sales) =SUMIF(SalesData[Month], "Sep", SalesData[Sale Price]) → Returns: $2,508.20 (September — highest month) General pattern: =SUMIF(SalesData[Month], [Month Abbrev], SalesData[Sale Price])
```

5.7 COUNTIF Formulas — Transaction Counting

```
-- Count transactions per product: =COUNTIF(SalesData[Product Description], "Water Pump") -- Count transactions per state: =COUNTIF(SalesData[Sale Location], "AZ") -- Count transactions per sales rep (first name match): =COUNTIF(SalesData[First Name], "Doug") -- Duplicate check: =COUNTIF($B$2:$B$172, B2) ← returns >1 if duplicate exists
```

5.8 AVERAGEIF Formula — Average Sale by State

```
=AVERAGEIF(SalesData[Sale Location], "CA", SalesData[Sale Price]) → Returns average sale price for California transactions General pattern: =AVERAGEIF(SalesData[Sale Location], [State], SalesData[Sale Price])
```

5.9 PivotTable Calculated Field — Profit Margin %

Inside the PivotTable, a Calculated Field was added for dynamic profit margin per filter selection:

```
Field Name: Profit Margin % Formula: = Profit / 'Sale Price' Steps: 1. Click inside PivotTable 2. PivotTable Analyze → Fields, Items & Sets → Calculated Field 3. Name: "Profit Margin %" 4. Formula: = Profit / 'Sale Price' 5. Click Add → OK 6. Format the field as Percentage with 2 decimal places
```

5.10 Unit Margin Formula (Products Sheet)

```
Unit Margin % = (Sale Price - Store Cost) / Sale Price × 100 Example — AutoVac: = (124.00 - 60.00) / 124.00 = 64 / 124 = 0.5161... → 51.6% In Excel cell: =(F2-E2)/F2 ← formatted as Percentage
```

6. Step-by-Step Dashboard Build Guide

This section documents the complete process used to build the **Profit_Tracker.xlsx** dashboard from scratch, starting from the clean dataset.

Phase 1 — Set Up the Workbook

- Open Data_Set.xlsx and copy all data (Ctrl+A → Ctrl+C)
- Create a new workbook: File → New → Blank Workbook
- Rename Sheet1 to 'DATA' — right-click tab → Rename
- Paste the data into the DATA sheet (Ctrl+V)
- Select all data including headers (A1:K172)
- Convert to Table: Insert → Table → check 'My table has headers' → OK
- Rename the Table to 'SalesData': Table Design tab → Table Name field
- Add a second sheet named 'DASHBOARD': right-click tab → Insert → Worksheet

Phase 2 — Create PivotTables

Six PivotTables were created as the data engine behind each chart. Each was placed on a hidden helper sheet named 'PIVOTS':

```
PivotTable 1 — Monthly Sales Trend: Source: SalesData table Rows: Month (sorted Jan→Dec) Values: Sum of Sale Price, Sum of Profit PivotTable 2 — Profit Per City (State): Source: SalesData table Rows: Sale Location Values: Sum of Profit PivotTable 3 — Commission Per Product: Source: SalesData table Rows: Product Description Values: Sum of Commision 10% PivotTable 4 — Sales Per Product: Source: SalesData table Rows: Product Description Values: Sum of Sale Price PivotTable 5 — KPI Summary: Source: SalesData table Values: Sum of Store Cost, Sum of Sale Price, Average of Sale Price, Sum of Profit PivotTable 6 — Top 5 Products: Source: SalesData table Rows: Product Description Values: Sum of Profit (sorted descending, Top 5 Value Filter applied)
```

Phase 3 — Sort Months Chronologically

Excel sorts months alphabetically by default (Apr, Aug, Dec…). To sort Jan–Dec chronologically:

- Click any cell in the Month row labels of the PivotTable
- Right-click → Sort → More Sort Options
- Select Manual sort
- Drag months into correct order: Jan, Feb, Mar, Apr, May, Jun, Jul, Aug, Sep, Oct, Nov, Dec
- Alternatively: File → Options → Advanced → Edit Custom Lists → add Jan,Feb,Mar... list
- With a custom list defined, PivotTable will auto-sort months correctly

Phase 4 — Build Charts from PivotTables

Chart	PivotTable Source	Chart Type	Key Settings
Monthly Trend	PT1 — Monthly Sales	Line with Markers	X-axis: Month, Y-axis: Sale Price
Profit Per City	PT2 — By State	Pie Chart	Data labels: State + $ amount
Commission Per Product	PT3 — Commission	Clustered Bar	Sorted descending by value
Sales Per Product	PT4 — Sales	Clustered Bar	Sorted descending by value
Top 5 Products	PT6 — Top 5	Horizontal Bar	Top 5 value filter applied

Phase 5 — Add the Month Slicer

- Click inside any PivotTable
- Go to PivotTable Analyze tab → Insert Slicer
- Check the 'Month' field → click OK
- A floating slicer panel appears — resize and position it top-left of the dashboard
- Connect slicer to ALL PivotTables: right-click slicer → Report Connections
- Check all 6 PivotTables in the dialog → OK
- Now clicking any month in the slicer updates all charts and KPIs simultaneously

Phase 6 — Build KPI Cards

KPI cards are simple cells formatted to look like dashboard tiles. For each KPI:

```
1. Draw a rectangle (Insert → Shapes → Rectangle) 2. Set fill to dark blue (#1F4E79), no border 3. Add two text boxes on top: - Label text (e.g. "TOTAL SALES") — white, 9pt, Helvetica - Value text linked to PivotTable cell (e.g. =PIVOTS!B5) — white, 18pt bold 4. Group the shape and text boxes (Ctrl+select all → right-click → Group) 5. Repeat for all 5 KPIs: Total Cost, Total Sales, Avg Sales, Total Profit, Profit Margin
```

Phase 7 — Dashboard Layout & Formatting

- Set the DASHBOARD sheet background to dark blue (Page Layout → Background, or fill all cells)
- Disable gridlines: View tab → uncheck Gridlines
- Hide row/column headers: View tab → uncheck Headings
- Add the title 'REVENUE, PROFIT AND COMMISSION TRACKER' as a large text box at the top
- Arrange KPI cards in a single row below the title
- Position the Monthly Trend and Profit Per City charts side by side in the middle row
- Place Commission Per Product and Sales Per Product charts in the lower row
- Position Top 5 Products as a vertical bar on the right side
- Move the Month Slicer to the top-left panel
- Remove chart borders, set chart areas to transparent to blend into dashboard

Phase 8 — Final Checks

- Test the slicer: click each month individually and verify all charts and KPIs update

- Select multiple months (Ctrl+click) and verify aggregated values are correct
- Click the clear filter button on the slicer and verify full-year totals match expected KPIs
- Verify Profit Margin = Total Profit / Total Sales = $6,356.70 / $17,110.60 = 37.15%
- Save the file as Profit_Tracker.xlsx
- Take a screenshot of the dashboard and save as Dashboard_Screenshot.png

7. Dashboard Components

Component	Type	Data Source	Insight Provided
Month Slicer	Slicer	Month column	Filter all visuals by one or more months
Total Cost Price	KPI Card	PivotTable — SUM	Total wholesale cost for selected period
Total Sales	KPI Card	PivotTable — SUM	Total retail revenue for selected period
Average Sales	KPI Card	PivotTable — AVERAGE	Mean transaction value for selected period
Total Profit	KPI Card	PivotTable — SUM	Gross profit for selected period
Profit Margin	KPI Card	Calculated Field	Profit as % of revenue for selected period
Monthly Trend	Line Chart	PT1 — Monthly	Revenue trend and seasonality over 12 months
Profit Per City	Pie Chart	PT2 — By State	Geographic profit share by US state
Commission Per Product	Bar Chart	PT3 — Commission	Commission cost breakdown by product
Sales Per Product	Bar Chart	PT4 — Sales	Revenue contribution per product
Top 5 Products	Bar Chart	PT6 — Top 5	Highest profit-generating products ranked

8. Analysis Findings & Insights

8.1 Top Products by Total Profit

Rank	Product	Total Profit	% of Total Profit	Transactions
1	Water Pump	$3,160.00	49.7%	20
2	AutoVac	$1,024.00	16.1%	16
3	Pool Cover	$842.10	13.2%	21
4	Skimmer	$630.00	9.9%	15
5	5 Gal Chlorine	$315.00	5.0%	9
6	Chlorine Test Kit	$150.00	2.4%	30
7	Net	$141.10	2.2%	29
8	Algea Killer 8 oz	$75.00	1.2%	15
9	8 ft Hose	$93.00	1.5%	31
10	1 Gal Muratic Acid	$18.00	0.3%	6

Key Insight: The Water Pump alone contributes nearly **50% of total profit** despite having a relatively lower unit margin (31.5%). This is driven entirely by its high unit profit value of $158 and consistent transaction volume.

AutoVac has the highest unit margin (51.6%) and is the second most profitable product.

8.2 Geographic Profit Distribution

State	Total Profit	Share	Total Sales	Margin
Arizona (AZ)	$1,979.10	31.1%	~$5,330	37.1%
Nevada (NV)	$1,710.10	26.9%	~$4,603	37.2%
California (CA)	$1,485.10	23.4%	~$3,996	37.2%
New Mexico (NM)	$681.80	10.7%	~$1,835	37.1%
Utah (UT)	$476.80	7.5%	~$1,283	37.2%
Colorado (CO)	$23.80	0.4%	~$64	37.2%

Key Insight: Arizona is the dominant market at 31.1% of profit. Colorado has negligible activity (0.4%) — representing either a new territory or one that needs attention. Profit margins are consistent (~37%) across all states, indicating uniform pricing with no geographic discounting.

8.3 Monthly Sales Trend

Month	Total Sales	Total Profit	Margin	Trend
January	$1,409.50	$523.60	37.1%	Strong start
February	$841.40	$312.60	37.2%	Decline
March	$1,421.40	$527.80	37.1%	Recovery
April	$1,089.70	$404.90	37.2%	Moderate
May	$214.40	$79.60	37.1%	Sharp drop
June	$409.90	$152.30	37.2%	Partial recovery
July	$186.10	$69.10	37.1%	Lowest month
August	$2,253.00	$837.20	37.2%	Strong surge
September	$2,508.20	$932.00	37.2%	Peak month
October	$1,927.60	$716.40	37.2%	High
November	$2,473.80	$919.50	37.2%	Near peak
December	$2,375.60	$882.70	37.2%	Strong close

Key Insight: There is a pronounced summer slump (May–July), with July being the lowest revenue month at only $186.10. This is counter-intuitive for a pool supply business (peak summer season) and may indicate either data limitations for that period or a business model driven by pre-season bulk orders (Jan–Apr) and post-season maintenance purchases (Aug–Dec). September is the single best month at $2,508.20.

8.4 Commission Analysis

Product	Total Commission	Total Sales	Commission Rate
Water Pump	$1,004.00	$10,040.00	10.0%
AutoVac	$204.80	$2,048.00	10.0%
Pool Cover	$168.42	$1,684.20	10.0%
Skimmer	$126.00	$1,260.00	10.0%
5 Gal Chlorine	$63.00	$630.00	10.0%
Chlorine Test Kit	$24.00	$240.00	10.0%
Net	$28.22	$282.20	10.0%
8 ft Hose	$27.90	$279.00	10.0%
Algea Killer 8 oz	$21.00	$210.00	10.0%
1 Gal Muratic Acid	$4.20	$42.00	10.0%

Total Commission Paid: $1,711.06 — representing 10.0% of $17,110.60 total sales. Commission is a flat rate with no tiered or performance-based variation in this dataset.

9. Setup & Usage Guide

9.1 System Requirements

Requirement	Minimum	Recommended
Software	Microsoft Excel 2016	Microsoft Excel 2019 / 365
Alternative	LibreOffice Calc 7.x	N/A — Excel preferred for full slicer support
OS	Windows 7 / macOS 10.12	Windows 10/11 or macOS 12+
RAM	4 GB	8 GB
Storage	10 MB free	50 MB free

9.2 Quick Start

- 1. Clone or download this repository (Code → Download ZIP on GitHub)
- 2. Extract the ZIP file to a local folder
- 3. Open **Data_Set.xlsx** first — review the raw data to understand the structure
- 4. Open **Profit_Tracker.xlsx** — the dashboard loads automatically
- 5. If prompted about external links or macros, click 'Enable Content'
- 6. Use the **Month slicer** (top-left panel) to filter by month
- 7. Click a single month to filter to that month; hold Ctrl to select multiple months
- 8. Click the clear button (funnel with X icon) on the slicer to reset to full-year view
- 9. All KPI cards, charts, and tables update dynamically with each filter change

9.3 Adding New Data

- Open Data_Set.xlsx and navigate to the DATA sheet
- Scroll to the last row of data (row 172) and add new transaction rows below
- Ensure all 11 columns are filled — Transaction Number must be unique and sequential
- Save Data_Set.xlsx
- Open Profit_Tracker.xlsx and press Ctrl+Alt+F5 (Refresh All PivotTables)
- All charts and KPIs will update to include the new data
- If using a named Table (SalesData), the range expands automatically — no manual range update needed

10. Technical Notes & Assumptions

Aspect	Detail
Commission Rate	Fixed at 10% of Sale Price for all reps and all products
Profit Calculation	Gross profit only (Sale Price - Store Cost). No overhead, tax, or shipping deducted

Aspect	Detail
Currency	All values in USD. No currency conversion required
Date Granularity	Monthly only. No day-level dates are recorded in the dataset
Floating-Point Precision	Minor floating-point artifacts (e.g. 4.9000000000000004) exist in raw data. ROUND(x,2) applied to commission column to resolve
Month Sort Order	Months are sorted chronologically (Jan–Dec) via a custom Excel list or manual PivotTable sort
Excel Version	Slicer + PivotChart functionality requires Excel 2013 or later
Data Scope	Dataset is fictional / educational. Not real commercial data
Product Codes	Each product has a unique numeric code. The same code always maps to the same product
Geographic Scope	US Southwest states only: AZ, CA, CO, NM, NV, UT

11. Extending the Project

11.1 Suggested Enhancements

- **Sales Rep Performance Dashboard** — Add a PivotTable sliced by First Name + Last Name to compare rep performance on profit and commission
- **Year-over-Year Comparison** — Extend the dataset with a Year column to track growth across multiple years
- **Dynamic Pricing Analysis** — Add a column tracking list price vs actual price to detect discounting patterns
- **Customer Segmentation** — Add a customer ID column to enable customer-level lifetime value analysis
- **Power BI Migration** — The same SalesData table can be imported directly into Power BI for richer interactive visuals and DAX measures
- **Python Analysis** — Load Data_Set.xlsx with pandas for statistical analysis, matplotlib charts, and automated reporting

11.2 Python Quick-Start (pandas)

```
import pandas as pd df = pd.read_excel('Data_Set.xlsx') # KPIs print(f"Total Sales: ${df['Sale Price'].sum():,.2f}") print(f"Total Profit: ${df['Profit'].sum():,.2f}") print(f"Profit Margin: {df['Profit'].sum()/df['Sale Price'].sum()*100:.2f}%") # Top products print(df.groupby('Product Description')['Profit'].sum() .sort_values(ascending=False).head(5)) # Monthly trend print(df.groupby('Month')['Sale Price'].sum())
```

12. License

This project is released under the **MIT License**.

You are free to fork this repository, adapt the dashboard for your own business data, and use the documentation as a template for your own data analysis projects.